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WOMEN SAFETY APP- SOS & LIVET RACKING

Database Systems Engineering and Distributed Backend

Abstract

- A safety system that provides a one-tap SOS alert and continuous live GPS tracking.
- Sends the user's real-time location and emergency alert to pre-selected trusted contacts.
- Outcome: faster emergency communication, continuous location visibility, and simpler use under stress.

Problem Statement & Objectives

- Problem: During emergencies, manual calls can be slow and difficult while under panic.
- Objective 1: Trigger an instant SOS alert with one tap.
- Objective 2: Continuously share live GPS location until the situation is resolved.
- Objective 3: Manage trusted emergency contacts securely and reliably.
- Scope: SOS, live tracking, contacts, alerts, backend and location data.

Existing vs Proposed System

Feature	Existing System	Proposed System
Alert	Manual call/message	One-tap SOS
Location	Often one-time sharing	Continuous live GPS
Interaction	Multiple steps	Minimal interaction
Connectivity	Often depends on stable network	Initial SOS prioritized
Gap	Separate tools	SOS + tracking + contacts in one flow

Feasibility Study

- Technical: Smartphone GPS, frontend, backend REST APIs, database and notification services.
- Economic: Uses commonly available/open-source development tools and software.
- Operational: Simple interface designed for minimal interaction during emergencies.
- Schedule: Modules developed and integrated within the semester project timeline.

Results & Testing

- SOS alert and location captured successfully.
- Trusted contacts receive the emergency alert and live-location link.
- Tracking remains active at regular intervals.
- Weak connectivity prioritizes the initial SOS alert.

Conclusion

- Provides instant SOS alerting with continuous live location tracking.
- Reduces interaction needed during an emergency.
- Keeps trusted contacts informed until the situation is resolved.

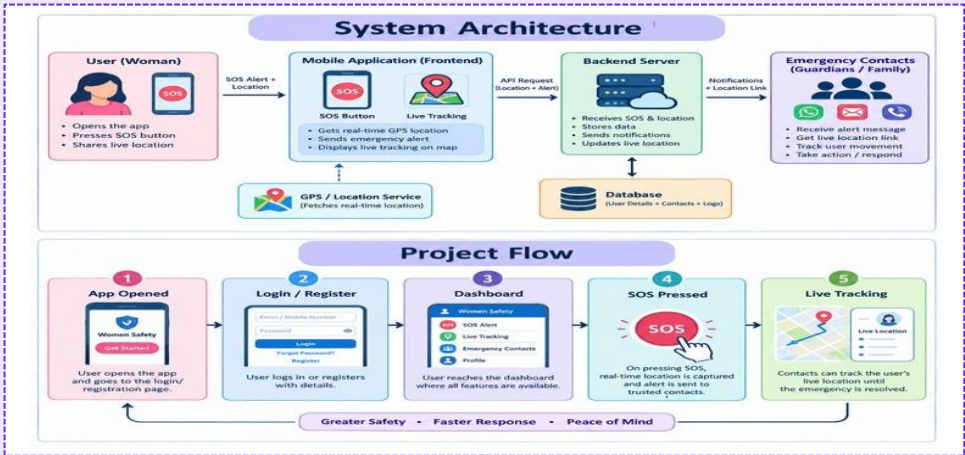
Future Scope

- Voice-activated SOS triggers
- Wearable / smartwatch integration
- Offline alert queuing
- Machine-learning-based anomaly detection
- Integration with local emergency services

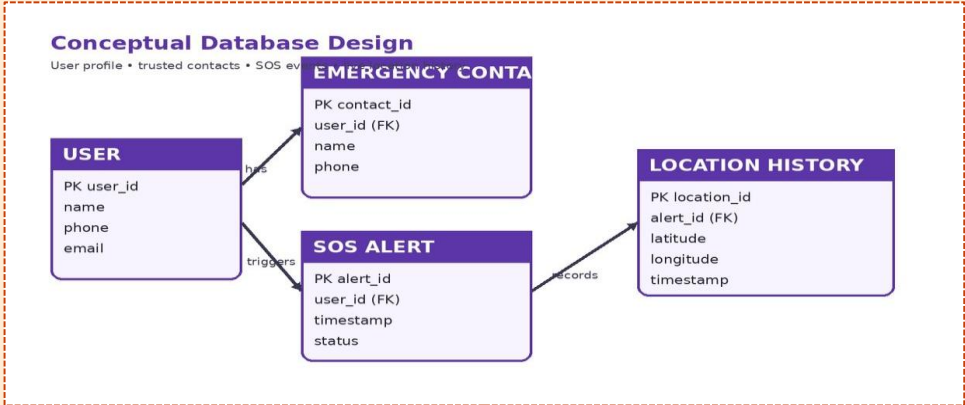
Tools & Technologies

- Front-End: React / HTML-CSS-JS
- Back-End: Node.js + REST APIs
- Database: MySQL
- Location & Maps: GPS + Google Maps API
- Notifications: Firebase Cloud Messaging
- Tools: VS Code, Git & GitHub

System Architecture



ER Diagram & Database Design



Implementation Screenshots

